ELECTRONICS · CHART 1 ESP32-C3

HazardGuard C3

One small board that watches four dangers at home or at work

Gas · Vapour · Heat · Water → Green · Yellow · Red + your phone

1 The problem

- LPG leaks, kitchen fires and burst pipes cause accidents every year.
- Most cheap alarms watch only **one** danger.
- You cannot check them from another room.
- Unplug a sensor and they still say 'all clear'.

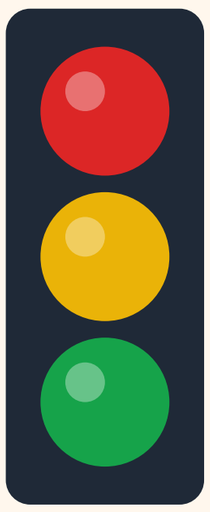
2 Our idea

- One ESP32-C3 watches **four** dangers at once.
- Three LEDs work like a **traffic light**.
- Its own WiFi: any phone opens a live dashboard.
- It tells you when a sensor is missing.

3 Four dangers, one board

Danger	Sensor	How it is noticed
LPG / gas leak	MQ-5	gas level rises above the limit
Alcohol / solvent vapour	MQ-3	vapour level rises
Overheating / fire	DHT11	temperature above 40 / 52 °C
Water leak / flooding	Soil probe	floor gets wet

4 Traffic-light warning

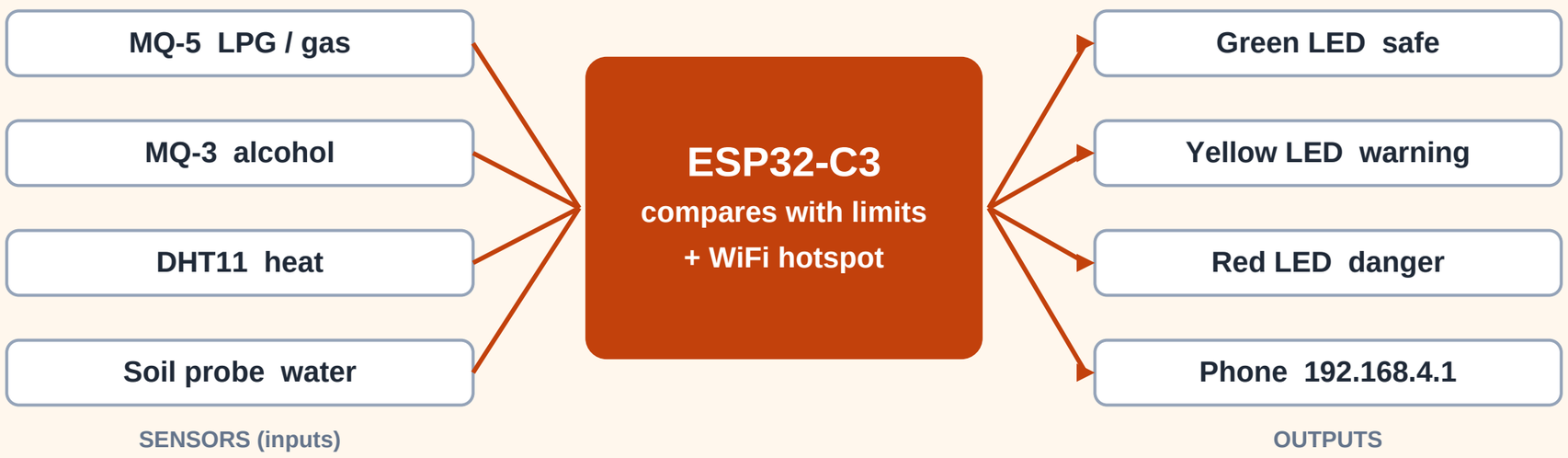


RED strobe
DANGER - act now
gas leak, vapour, fire or flooding

YELLOW blink
WARNING - check
a reading is rising

GREEN steady
SAFE
all sensors normal

5 Parts we used

Part	Job	Rs.
ESP32-C3 SuperMini	brain + WiFi hotspot	300
MQ-5 gas sensor	LPG / natural gas	150
MQ-3 gas sensor	alcohol, solvents	150
DHT11	temperature, humidity	110
Soil moisture probe	water on the floor	80
3 LEDs + 220 Ω	green / yellow / red	15
5 V USB adapter	power	150
Total		955

6 How the parts connect